

Lopsided thin products and the IC pipeline: integration report (2026-10-06)

Question: can the Alman–Vassilevska Williams thin-matrix-product technique ([arXiv:2610.06783v1](https://arxiv.org/abs/2610.06783v1)) reduce end-to-end index-calculus cost in this repository? Scope, method, frozen inputs, and stop rules are in the accompanying protocol (README.md). Statement status is labelled **[proposed]** / **[derived]** / **[verified]** / **[measured]** (none in this round).

Method

1. Downloaded [the paper PDF](#) (v1, 76 pages; sha256:d91a79c5dbf1fc662f0c5c1bb2b690ac3d83d9ef5bf872bde4cdaf7cc9e76005) **[verified]**; read Sections 1–2, 4 (core technique) plus 3, 5.2 (reductions) at survey depth.
2. Implemented the problem interface natively in Rust (src/cryptanalysis/lopsided_thin_product.rs, 8 unit tests) **[verified]** (cargo test --lib cryptanalysis::lopsided_thin_product: 8 passed).
3. Assessed four IC insertion points; implemented none **[derived]**.

What the paper shows **[derived]**

- **Thin product (Thm 1 / Thm 5).** $X: N \times D, Y: D \times N, D \leq N^{(1/18)}, |W| \leq N^2/\sqrt{D}$ wanted positions: all $(XY)[i, j]$ for $(i, j) \in W$ in $O(N^2/D^{0.063})$ operations — polynomially less than one operation per entry of the full product. General form: for every $\varepsilon < 0.1204, \kappa > 0$ there is $\gamma > 0$ with $O(N^2/D^\gamma)$ when $D \leq N^\varepsilon, |W| \leq N^2/D^\kappa$.
- **Technique.** Schoenhage's 10-multiplication identity inside a Coppersmith-style recursion, visiting only the recursion-tree leaves the wanted entries need; encoding work is shared across blocks.
- **Graph reading.** With X, Y as biadjacency matrices this is the counting Lopsided All-Edges Sparse Triangle: $(XY)[a, b]$ counts middle vertices adjacent to both outer vertices a, b . Cost for $\varepsilon < 1/18$: $O(|W| n^{(0.437 - \varepsilon)} + n^{(2 - 0.063 - \varepsilon)})$.
- **Data structure (Thm 3).** Preprocess (X, Y) in $O(N^2/D^{0.063})$, answer any single entry in $O(D^{0.437})$.
- **Reductions.** 3SUM / Exact Triangle / APSP reduce to the lopsided problem, giving $O(n^{1.9992})$ 3SUM and $O(n^{2.9995})$ APSP. SETH / Orthogonal Vectors are explicitly unaffected.

Pipeline diagram

Thin lopsided products into the IC pipeline — proved technique vs this round's proposals

Source: Alman–Vassilevska Williams, arXiv:2610.06783v1. Green = proved in the paper. Amber = proposal assessed (not implemented) in this round.

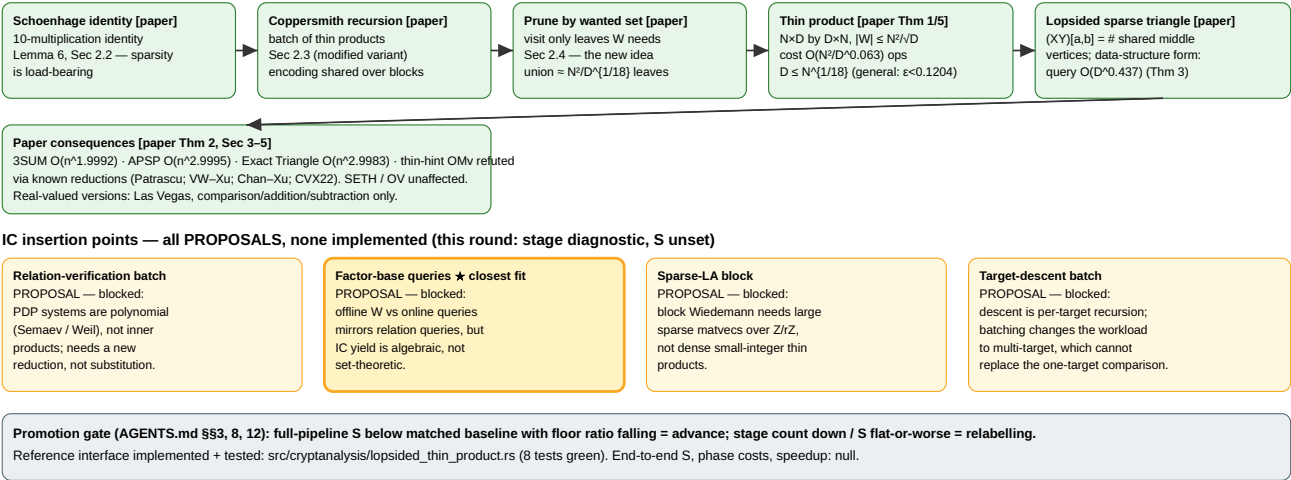


Figure 1 — green boxes are proved in the paper; amber boxes are proposals assessed (not implemented) in this round. The ★ box is the closest fit and still blocked. Editable source: figure.mmd.

Findings for our IC pipeline

Insertion point	Implemented?	Blocking difference [derived]	Required experiment [proposed]
Relation-verification batch	No	PDP systems are polynomial (Semaev/Weil), not inner products; needs a new reduction	Matched baseline/candidate full-DLP suites, all phases in S
Factor-base intersection query	No — closest fit	Offline w vs online queries mirrors relation queries, but IC yield is algebraic, not set-theoretic	One-target workloads paired vs matched rho, five online phases named
Sparse-LA block	No	Block Wiedemann needs large sparse matvecs over Z/rZ , not dense small-integer thin products	Price matrix build + final LA inside total S
Target-descent batch	No	Descent is per-target recursion; batching changes the workload to multi-target	Single-target first, then a separate k-target question

Positive **[verified]**: the interface (spec validation, regime gates, epsilon/kappa, checked naive reference, quoted-bound hooks, Boolean triangle-count equivalence) is implemented and tested.

Negative **[verified]**: no insertion point is wired into any candidate; S, phase costs, and speedup are null; no ecbench session was run; no curve was named or registered.

Cost-model illustration (quoted bounds, not measurements)

Quoted-bound illustration — NOT a measurement (log scale, operations, N = 2^36)
Curves evaluate closed forms from arXiv:2610.06783v1: naive-max = $N^2\sqrt{D}$ ($|W| = N^2/\sqrt{D}$ inner products) · full $\approx N^2$ · paper = $N^2/D^{0.063}$ (Thm 1). No code was timed.

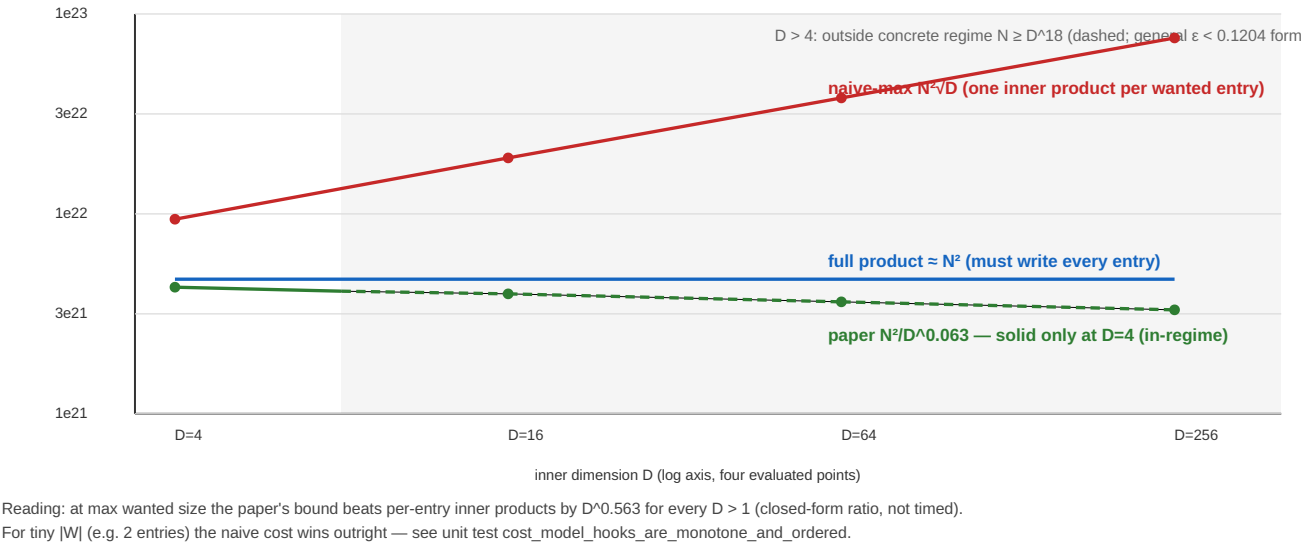


Figure 2 — closed-form evaluation of the paper's bounds at $N = 2^{36}$. At max wanted size the paper's bound beats per-entry inner products by $D^{0.563}$ for every $D > 1$ (closed-form ratio, not timed). For tiny $|W|$ the naive cost wins outright. Only $D = 4$ satisfies the concrete-regime gate $N \geq D^{18}$ at this N .

Boundary, table, ratio (AGENTS.md §§1–8)

- Boundaries stated before measuring: generic floor + matched rho, both unchanged; the unit is S.
- No table row is produced: a row without a verified answer is not a result.
- Falsification target for a follow-up: full-pipeline S below the matched baseline with the floor ratio falling (advance); S down with the floor ratio flat is engineering; a stage count down with S up is relabelling; a re-derived number with no algorithm change is accounting.
- All phases unpriced (null); wall time appears nowhere.

Graphs checked — no change [verified by inspection]

- docs/index-calculus-scoreboard.html + docs/ic/progress-timeline.json: no new IC/rho ratio, no point added.
- docs/ic-leaderboard.html / LEADERBOARD.md / leaderboard.json: no whole-pipeline measurement.
- docs/browser/data.json: no new curve, session, round, or IC1 identity.
- docs/curves/registry.json: no new curve named; no retired spellings introduced.
- Figures 1–2 above are new visuals for this search round (research-visuals skill), not canonical-graph updates.

Open checks [proposed]

1. A follow-up protocol implementing the pruned-recursion algorithm natively and replaying it against `wanted_product_i64` on frozen inputs.
2. Whether factor-base intersection queries on any concrete toy base admit a thin-product encoding that preserves decomposition probability — reduction argument first, code second.
3. Re-reading the paper's Section 4 trade-offs once a concrete IC shape $(N, D, |W|)$ is proposed.

Sources

- Paper PDF: <https://arxiv.org/pdf/2610.06783> (v1).
- Code: `src/cryptanalysis/lopsided_thin_product.rs` (this repo, this PR).
- Visuals: `figure.mmd` / `figure.svg`, `cost_model.svg`.